Predicting USD/LKR Exchange Rate: A CRISP-DM Machine Learning Approach

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Abstract – Dollar dominance a well-known word which describes the power of the dollar. Most of the countries around the world depend on the dollar for imports and exports. Depth from the IMF and the World Bank have been provided by US dollar. This study shows how the dollar became so dominant, and what are the steps to be taken for developing countries like Sri Lanka to minimize the burden of the dollar. A CRISP-DM model has been made to analyze the fluctuations of the dollar rate in Sri Lanka and predict the dollar fluctuation using Random Forest, XGBooost and Ensemble algorithms. This study speaks about the history of the dominance of currency, how dollar dominance affected Sri Lanka, what are the main reasons for the downfall of the Sri Lankan economy are all diccussed in this paper.

Keywords*: Dollar dominance, petrodollar, GDP (Gross Domestic Product), Federal Reserve, FX reserves*

I. INTRODUCTION

The power of the currency depends on certain factors such as the Economic strength- Size of the economy and the trade volume, Market development and openness, price stability. Currently, the US Dollar is the most relied currency by the world, except for countries such as Russia and China. In the 1600s the most dominating currency was the Dutch guilder, when the Netherlands became the major trade and finance center. This was during the renaissance where the Dutch explored in many fields such as banking and finance. In 1609 they opened the bank of Amsterdam where they used to preserve gold and issue currency. In the 1800s where the Industrial revolution begins, trades needed an alternative for the guilder. The Industrial revolution, which was led by Britain, quickly became an economic powerhouse, due to the usage of steam power and other innovations, which made the country the largest economy and commercial power, this eventually made the Sterlin pound more dominant than the guilder. It is notable that Netherlands did not

have the largest economy even in the 1600s. The British further had an open and well-developed financial system, which make the pound more dominant. The pound was the most dominant currency for more than a century. Back-to-back world wars made many destruction all around the world leaving America intact, Since Second World War the US dollar played a vital role as a currency of global reserve [1] This gave the US economy an advantage and traders begin to show more importance on the Dollar.

In 1944, The Bretton Woods Agreement made a new international monetary system where the economic system center around the dollar, foreign countries agreed to peg their currencies to the US Dollar while the US Dollar was pegged to gold. The Bretton Woods Agreement which held in July 1944 was conducted by approximately 730 delegates representing 44 countries with a goal of creating an efficient foreign exchange system, preventing competitive devaluation of currencies and promoting international economic growth. The agreement created two important organizations, the IMF (International Monetary fund) and the World Bank. The Bretton Woods Agreement ended in 1971, but the IMF and the World Bank remained as a strong pillar for international currency exchange. Even though the agreement ended, The United States has the most dominant currency – the US Dollar up to date due to reasons such as maintain the worlds largest economy, connection among world's largest traders, an open and efficient financial market, and the most safe and stable asset for the US debt. In simple terms the US economy is the open, most innovative and the most productive which keeps the US Dollar the most convenient, liquid and reliable currency among all other international currencies. The Dollar isn't the perfect currency, but for trades and investors it's the least bad option, China trade more than the US but in has restrictive financial markets, The United Kingdom has a highly open market, but still trades much less than the US, the Euro and Japanese Yen also have certain limitations. The US Dollar has the advantage of connection among largest traders and the network created by the flow of dollar among the world, this makes the Dollar more dominant and not easy to replace.

As of 2025, the dollar remains dominant store of value in foreign exchange reserves accounting up to 58% of global reserves, and 88% of foreign exchange transactions and serves as invoicing currency for 80% of global trade [2]. These shows that the dollar becomes a global financial anchor and acts as a bridge among foreign currency exchange, while concentrating economic and geopolitical power in the United States of America.

The Dollar Dominance has a huge effect on many developing countries. The fluctuation in the dollar rate affects the day today living of citizens due to the dependency of the dollar. Sorting out the depths of the country have also been a big challenge for those countries as for the change of interest rate, when repaying the depths, it increases by twice or thrice. The cost of importing and exporting of goods also depend on dollar fluctuations, which weakens the local currency rate. For developing economies, the dominance of the dollar has major difficulties about dependency in US monetary policy, currency mismatch in national balance sheets and limitations for policy autonomy. One such developing country is Sri Lanka, which has its own production markets like Tea and geographical locations but depends on exports such as crude oil, medicine, industrial raw material and vehicles, has a significant impact. The country is liable for depths in the IMF, the easter attack, the cost of import and political constraints have been a very challenging task for the citizens in the country when the dollar has a significant fluctuation.

In the 2022 crisis, Sri Lankas foreign exchange reserves fell from USD 7.6 billion in 2019 to USD 20 million by 2022 [2], which means the country did not have enough dollars for

importing products such as fuel, medicine and even for dept repayments. This crisis how dollar dominated trade and dept exposed for developing economies to sudden debt distress, inflation explosion, social instability when access for foreign currency was disrupted. Global public depth rose to USD 102 trillion in 2024, developing countries accounted for one third of that amount USD 31 trillion and paid USD 921 billion in interests [3].

This research carries four objectives, 1. Developing an accurate USD/LKR exchange rate. 2. Implement CRISP-DM framework for a systematic model development. 3. compare performance of the statistical and machine learning. Further, analyze dollar dominance patterns in Sri Lanka, identify key economic indicators affecting exchange rates and provide actionable insights for policymakers. The primary objective of research is to address on to what extent does USD in global trade and energy influence exchange rate fluctuates, and how and why the dollar dominance has a major effect on Sri Lanka.

HISTORIC EVOLUTION ON DOLLAR DOMINANCE

The Bretton Woods Agreements which took place in 1944 was the beginning of dominating the dollar, even those the agreement ended in the 1970s the US made several strategies to maintain the dollar dominance.

In 1971 President Richard Nixon address the Nixon shock openly in the Whitehouse, which highlighted the importance of maintain the dollars position as a stable monetary value and pointed aims for defending the currency, he further mentioned about supporting for American Jobs and Promoting a fair trade internationally. He mentioned that “I have directed secretary Cornelly suspend temporarily the convertibility of the dollar to gold and other reserve assets except to be in the interest monetary stability and in the best interest of United States”. [4]

The petrodollar agreement which was officially signed on the 8th of June 1974 was a major reason for US global economic influence. This agreement was signed with the Saudi Arabia, which meant that crude oil will be sold in US Dollars for exports, while Saudi Arabia receives military and equipment. This was a 50-year agreement which ended on the 9th of June 2024. The petrodollar agreement was a main reason for the dominance of the dollar rate as people from other countries including Sri Lanka should export crude oil using the dollar. Even though the agreement ended on the June 2024, till today people exchange crude oil using the dollar, but the end of the agreement has become a threat to the US, and the Saudi Arabia did not want to extend to an 80-year agreement. But the petrodollar agreement is a significant cause for the development of the network of dollar usage, which has been very stable and not easy to be changed. The network effect brought by the petrodollar agreement not only increased the power of the dollar as a currency but also became the primary currency used to export crude oil. That deadline of US credit will threaten the US dollars currency status, which is reflected in the possible long term downward trend of U.S. dollar in the foreign exchange market [5].

IMPACT ON SRI LANKA CASE STUDY

Sri Lanka faced a lot of challenges last few years, from the easter attacks in 2019, then the impact of corona virus in 2020 and 2021, further from 2019 - 2022 they faced political issues, again in 2024 the had to go through the Ditwah floods. All these factors made Sri Lanka face more problems related to the country’s economy.

In 2019 after the easter attacks, tourisms which was the main source of the country’s income was hindered. The restriction of tourism then was a reason for Sri Lanka to make it more difficult to repay debts. This reduced the

countries GDP, which we will be discussing in this paper later. Tourism after the Easter Sunday attacks reduced by 18%, which made the earning brought by tourists to reduce by 17.7% to US Dollar 3607 million in 2019 [6].

2020/2021 COVID-19 Virus made the hugest impact on tourism, not only in Sri Lanka but all around the world. The restriction of tourism in this period and the implementation of lockdowns and curfew made it very difficult for the country to gain its necessary income. In this period the economy of Sri Lanka decreased by 3.6% in 2020, which was recorded the worst growth performance [7].

In 2019 the country had begun new leadership, there were few bad decisions by the then government such as a tax cut made the country suffer more as the country's revenue is built by tax. In 2022 the Country faced a political issue, this was due to the decision made by then leaders to over come these debts, but due to poor knowledge in implementing decisions as necessary. People gathered for protests in huge crowds which again made a reason to reduce the country's GDP, because the government couldn't make any decision and the citizen in the country were completely against every decision and even the whole leading party. By mid 2022, official reserves reduced drastically.

After all these Sri Lanka fell even more, not only the issues in Sri Lanka, but inflation rates in the United States which indirectly affected the debt in Sri Lanka made a big impact for the country.

Later In 2024, Sri Lanka had to face another milestone which was the Ditwah floods, many people lost lives, lost plantation which included rice, vegetable, fruits and tea which is the main source of export income got affected badly.

II. LITREATURE REVIEW

As we all know, the United States are also in depth. But why doesn't the depth in United States affect as the depths in developing countries such as Sri Lanka? It is Because the Federal Reserve, which is the central Bank of the United States can print dollars and the Central Bank of Sri Lanka prints Lankan rupees. Now the debts must be repaid by dollars for both the countries. If the United States needs to repay all its debts, they can get the dollars printed through the Federal Reserve and pay all the debts. But when it comes to Sri Lanka, if we print Sri Lankan rupees, it will weaken the rupee value even more. Now, why doesn't the Federal Reserve print the US dollars and settle all its depths. It is because, that if you keep printing more currency, the value of that currency would reduce. This would reduce the power of that currency. Therefore, the US does not have the main idea of repaying the depths, but still their focus is on maintaining the dollar dominance. So that the risk of the depths is less for the US. Since Sri Lanka depends on dollars for imports and debt repayments. There is a big impact for the Sri Lankan rupee through the US dollar. The knowledge of the relationship between US fluctuation and the exchange rate and import volume will support to gain a beneficial trade and be prepared for much stable situation within Sri Lanka [8].

The GDP, which is known as the Gross Domestic product is the total monetary value of all the final goods and services produced within a country during a specific period which is usually a year. GDP measures the size of the country's economy. The GDP is calculated as, $GDP = C + I + G + (X - M)$. Where C is consumption. I - Investment, G - Government Spending, X - Exports and I for imports. Currently, the United States has the highest GDP. China is competing The United States and currently has the highest amount of trade bypassing the trade in the United States.

From about 1978 to 2015 the country's yearly import demand shows over 30% share of GDP except 1984 2009 2010 2013 and 2015 [8].

The trade of the country is the main reason for the development of the economy of that country. US has a leading trade economy, but China has just bypast the due to their production in large quantity and sale. The international balance of payments that is changes in foreign exchange supply and demand costs by trade deficit and surplus due to exchange rate fluctuations [9]. Sri Lanka has a limited amount of trade like tea and garments, but it depends on medicine, vehicles, fuel and many more. This dependance increases the trade in other countries. From 2013 to 2014 about 8 USD millions and 2014 to 2015 about 1 USD million Sri Lanka had a wide trade deficit. This deficit is persisting over the years in Sri Lanka economy [8].

By introducing economic growth factors and optimal inflation model under more realistic general framework can be established [5]. The increase of US current account deficit and its foreign liabilities cannot be explained by the economic growth. The economic growth can be seen because of the dollar circulation [10].

Prediction of the foreign exchange rate can be considered an Important thing. That is because the exchange rate of a country. Plays a crucial role in its economic growth [11]. Previous studies show many models and algorithms which is used to understand the trend of the USD fluctuations with other currencies. Such as the use of a ARDL (Autoregressive Distributed Lag) approach to identify cointegration between US dollar index volatility and various import categories [8]. The contribution of inflation to debt relief and the drag of growth decline on debt relief mainly depend on the hypothesis of relationship between growth and inflation [5]. The main aim of this study is to analyze the impact of the dollar fluctuation and make a model understand the trends to predict the dollar rate in the near future using a CRISP-DM method, this study also focuses on policy making to protect the country's economy.

III. METHODOLOGY

The development process of data mining has been standardized and outlined by the Cross Industry Standard Process for Data Mining (CRISP-DM) [12]. This process has 6 steps, business understanding, data understanding, data preparation, modeling, evaluation and deployment.

BUSINESS UNDERSTANDING

The main objective is to develop an accurate forecasting model for the LKR/USD exchange rate with the following requirements.

1. **Target Accuracy**: Mean Absolute Percentage Error (MAPE) < 0.5%
2. **Explanatory Power**: R-squared (R^2) > 0.95
3. **Uncertainty Quantification**: 95% confidence intervals for risk assessment
4. **Forecast Horizon**: 30-day forward predictions

The model is expected to find solutions for what the exchange rate will be within the next 30 to 90 days for the temporal pattern and seasonality exists in the data set and the confidence level of the predictions.

DATA UNDERSTANDING

Data source and collection

The data set consist of daily exchange rates from Lankan rupees to U.S. dollars obtained from the Central Bank of Sri Lanka. From May 7th, 2010 to February 9th, 2026. The data set contains 3346 observations with date and exchange rate as its attributes.

Exploratory data analysis

Statistical analysis gives us the following characteristics in Table 1.

Table 1- Statistical Summary

	Date	Exchange Rate
count	3346	3346.000000
mean	2019-01-26 03:07:12.516437	201.102171
min	2010-05-07 00:00:00	109.459700
25%	2015-09-11 18:00:00	138.960625
50%	2019-03-03 00:00:00	176.539250
75%	2022-08-23 18:00:00	296.135900
max	2026-02-09 00:00:00	364.760000
std	NaN	78.547583

The augmented Dickey Fuller (ADF) test was done in order to understand the data stationarity. The results were as follows,

- ADF Statistic: -1.100127
- p-value: 0.715021
- Result: Non-stationary ($p > 0.05$)

Seasonal decomposition using additive model for a period of 30 days was performed to identify. Trend component (Long term movement showing a appreciation of USD against LKR.), Seasonal component (Monthly cyclic patterns) and Residual Component (Fluctuations after removing trend and seasonality).

DATA PREPARATION

Feature engineering

To analyze the change of the exchange rate 14 engineered features were constructed from the dataset. They were,

Moving Averages (Trend Features):

$$MA_7 = (1/7)\ \Sigma(t=i-6 \text{ to } i)\ Rate_t$$
$$MA_30 = (1/30)\ \Sigma(t=i-29 \text{ to } i)\ Rate_t$$
$$MA_90 = (1/90)\ \Sigma(t=i-89 \text{ to } i)\ Rate_t$$

Volatility Measures:

$$Volatility_7 = sqrt((1/7)\ \Sigma(t=i-6 \text{ to } i)\ (Rate_t - MA_7)^2)$$
$$Volatility_30 = sqrt((1/30)\ \Sigma(t=i-29 \text{ to } i)\ (Rate_t - MA_30)^2)$$

Range Indicator:

$$Range_30 = max(Rate_(i-29) \text{ to } Rate_i) - min(Rate_(i-29) \text{ to } Rate_i)$$

Lag Features:

$$Lag_k = Rate_(i-k), \text{ where } k \in \{1, 2, 3, 5, 7, 14, 30\}$$

These features made the model to capture short-term momentum, medium term trends, long term patterns and market volatility.

Data splitting

The data was divided into training and testing set. The training set consists of 80% of the data, which is 2605 samples with LKR/USD exchange rates from 23rd May 2011 to 29th of May 2023. The test set consist of 20% of the data, which is 652 samples with LKR/USD exchange rates from 30th of May 2023 to the 9th of February 2026.

Data splitting ensures that the model is evaluated on future unseen data. Which stimulates the real-world deployment conditions.

Data quality processing

Check for duplicate data entries, Excluded the currency column.

MODELING

Three machine learning models were implemented and compared. Which is followed by an ensemble approach.

Random forest regression

Random Forest which is a machine learning algorithm operates by building numerous decision trees throughout the process of training. And then produced a class that

represents the mean of the classes (Classification) and the prediction (Regression) of individual trees [13].

Algorithm Specification:

Algorithm: Random Forest
Parameters:
- n_estimators = 500 trees
- max_depth = 15
- min_samples_split = 5
- min_samples_leaf = 2
- random_state = 42
- n_jobs = -1 (parallel processing)

Mathematical Formulation:

For an input vector **x** with features x_1, x_2, ..., x_{13}, the Random Forest prediction is:

$$\hat{y} = (1/T) \Sigma(t=1 \text{ to } T) h_t(x)$$

where T = 500 trees and h_t(x) is the prediction of the t-th decision tree.

Advantages of using the Random Forest regressor includes robust to overfitting through averaging, captures nonlinear relationship, provides feature important rankings and Handles multicollinearity naturally.

XGBoost (Extreme Gradient Boosting)
XGBoost is a lifting tree model, it integrates many tree models to form a strong classifier. The model used is the classification and regression tree model [14].

Algorithm Specification:

Algorithm: XGBoost
Parameters:
- n_estimators = 500
- max_depth = 6
- learning_rate = 0.01
- subsample = 0.8
- colsample_bytree = 0.8
- random_state = 42

Objective Function:

XGBoost minimizes a regularized objective:

$$L(\varphi) = \Sigma(i=1 \text{ to } n) l(\hat{y}_i, y_i) + \Sigma(k=1 \text{ to } K) \Omega(f_k)$$

where:

- $l(\hat{y}_i, y_i)$ is the loss function (MSE for regression)
- $\Omega(f_k)$ is the regularization term controlling model complexity
- f_k represents the k-th tree

Regularization Term:

$$\Omega(f) = \gamma T + (1/2)\lambda \Sigma(j=1 \text{ to } T) w_j^2$$

where T is the number of leaves, w_j are leaf weights, γ and λ are regularization hyperparameters.

Advantages of XGBoost model is that it has a built in regularization which prevents over lifting, supports handling of sparse data, parallel tree construction and superior performance on structured and tabular data.

Gradient boosting regressor
Gradient boosting uses trees that are iteratively build up to optimize the second order tail expansion of the loss function. That includes gradients and Hessians to improve optimization [15].

Algorithm Specification:

Algorithm: Gradient Boosting
Parameters:
- n_estimators = 500
- max_depth = 5
- learning_rate = 0.01
- subsample = 0.8
- random_state = 42

Prediction Function:

$$F_m(x) = F_(m-1)(x) + \nu \cdot h_m(x)$$

where:

- $F_m(x)$ is the ensemble prediction at iteration m
- ν is the learning rate (0.01)
- $h_m(x)$ is the m-th weak learner fitting the negative gradient

Ensemble model.

An ensemble model combines the predictions from all the previous used models using weighted average to obtain their complementary strengths.

Ensemble Formula:

$$\hat{y}_{ensemble} = \Sigma_{(i=1 \text{ to } M)} w_i \cdot \hat{y}_i$$

$$\text{subject to: } \Sigma_{(i=1 \text{ to } M)} w_i = 1, w_i \geq 0$$

Weight Assignment:

Weights were assigned based on individual model performance on validation data:

$w_{XGBoost} = 0.40$ (best performer)
$w_{RandomForest} = 0.30$
$w_{GradientBoosting} = 0.30$

The model has diversity, robustness, stability and generalization.

EVALUATION

Four metrics were used to access model performance.

a) Mean Absolute Percentage Error (MAPE): Average percentage deviation from the actual values, lower is better.

$$MAPE = (100/n)\, \Sigma_{(i=1 \text{ to } n)} |y_i - \hat{y}_i| / |y_i|$$

b) R-Squared (Coefficient of Determination): Proportion of variance range from 0 to 1, higher is better

$$R^2 = 1 - (SS_{res} / SS_{tot})$$

where:

$$SS_{res} = \Sigma_{(i=1 \text{ to } n)} (y_i - \hat{y}_i)^2$$

$$SS_{tot} = \Sigma_{(i=1 \text{ to } n)} (y_i - \bar{y})^2$$

c) Mean Absolute Error (MAE): Average absolute prediction error in original unit LKR

$$MAE = (1/n)\, \Sigma_{(i=1 \text{ to } n)} |y_i - \hat{y}_i|$$

d) Root Mean Squared Error (RMSE): Penalizes larger errors more heavily than Mean Absolute Error

$$RMSE = \text{sqrt}((1/n)\, \Sigma_{(i=1 \text{ to } n)} (y_i - \hat{y}_i)^2)$$

Cross model comparison

A comparative analysis was performed across all models using the test set with 652 samples

Feature importance analysis

XGBoost provides feature importance course using gain metrics

$$Importance_j = \Sigma_{(i:\, j \in split(i))} Gain_i$$

where $Gain_i$ is the improvement in splitting criterion when using feature j.

DEPLOYMENT

Recursive multi step forecasting

A recursive forecasting strategy was implemented to obtain predictions which is 30 days ahead.

Algorithm: Recursive Forecasting

Input: Historical data $H = \{x_1, ..., x_t\}$, Models M
Output: Forecasts $F = \{\hat{y}_{(t+1)}, ..., \hat{y}_{(t+30)}\}$

For day d = 1 to 30:
1. Compute features from $H \cup \{\hat{y}_{(t+1)}, ..., \hat{y}_{(t+d-1)}\}$
2. For each model m in M:
   $\hat{y}_m$ = m.predict(features)
3. Ensemble prediction:
   $\hat{y}_{(t+d)} = \Sigma\, w_m \cdot \hat{y}_m$
4. Append $\hat{y}_{(t+d)}$ to history

Characteristics

Day 1 uses 100% historic data, day 2 to day 30

uses previous predictions into feature calculations, Moving averages likes and volatility metrics recalculated each step and the error propagation uncertainty increases with forecast horizon

Confidence interval estimation

95% confidence intervals were computed using empirical error distribution from the test set:

$$CI_95 = \hat{y} \pm 1.96 \cdot \sigma_test$$

$$\text{where } \sigma_test = \text{std}(\{y_i - \hat{y}_i\} \text{ for all i in test set})$$

Production implementation specifications

- **Programming Language**: Python 3.11
- **Core Libraries**:
  - scikit-learn 1.4.0 (Random Forest, Gradient Boosting)
  - XGBoost 2.0.3 (XGBoost)
  - pandas 2.1.4 (data manipulation)
  - numpy 1.26.3 (numerical operations)
- **Hardware Requirements**: Standard CPU (no GPU required)
- **Training Time**: ~3-5 minutes on consumer hardware
- **Inference Time**: <100ms for 30-day forecast
- **Model Persistence**: Serialized using pickle for deployment

IV. RESULTS

This section presents the complete evaluation of the processed and ensemble machine learning framework used for LKR/USD exchange rate forecasting. The results are organized into 5 subsections; model performance comparison, feature importance analysis, prediction accuracy visualization, error analysis and deployment forecasting.

HISTORICAL DATA ANALYSIS

Figure 1 presents a complete historic dataset for 16 years from 2010 to 2026 of daily LKR/USD exchange rate obtained from the Central Bank of Sri Lanka.

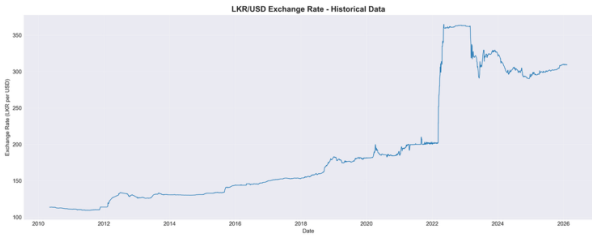

Figure 1 - Historical LKR/USD exchange rate data (2010-2026)

Figure 1 shows 3 distinct faces according to the time.

1. Stable Period (2010-2020)- Depreciation from 110 to 200 LKR USD highlights an annual depreciation rate of approximately 6.2%.
2. Crisis Period (2022)- A huge decrease to 365 LKR/USD representing an 80% depreciation within months this is due to the political effects which reduced taxes and implemented import restrictions.
3. Recovery Period (2023-2026)- stabilization around 300 to 310 LKR/USD following the IMF intervention and structural adjustment programs.

Temporal decomposition analysis

To understand the patterns seasonal decomposition was performed using an additive model within a 30 day. Results are presented in figure 2.

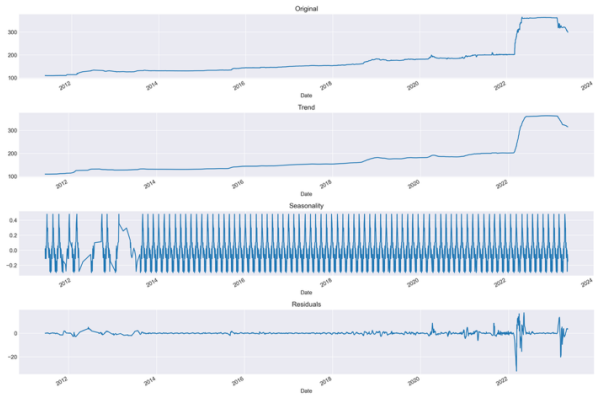

Figure 2 - Seasonal decomposition of LKR/USD exchange rate

Figure 2 illustrates the extracted trend smoothly capturing the long-term direction movement without short term fluctuations. The trend clearly identifies the 2022 crisis as an inflection point. Figure 2 also shows monthly cyclic patterns which are evident throughout the series. Further, it shows the residual represents random fluctuations after removing trends and seasonality here the residuals are near-zero during 2014 to 2021 the stable period. Large spikes appeared during 2022-2023 the crisis and recovery period and recent residuals (2024-2026) showed reduced magnitude indicating return to back to normal.

Model performance comparison

Table 2 presents a summary performance evaluation of all the models on the text set with 652 samples from may 2023 to February 2026

Table 2- Model Comparison (Sorted by MAPE)

	MAE	RMSE	MAPE	R^2
XGBoost	2.543190	3.320276	0.837958	0.904049
Ensemble	2.847095	3.578095	0.939873	0.888570
GradientBoosting	3.313092	3.897523	1.084811	0.867786
RandomForest	3.599276	4.490996	1.188341	0.824456

Table 2 shows that XGBoost achieved the best performance across all the metrics and we then MAPE of 0.838% which outperforms the 2% industry benchmark and with the R^2 of 0.904 indicates 90.4% of variance explained and RMSE/MA E ratio of 1.31 which shows the consistent prediction without outliers. Ignoring the slight low accuracy MAPE 0.940% the ensemble model provides a reduce overfitting risk, more stable predictions and better generalization. The performance gap is that XG boost outperforms random forests by 29.5% which shows the superiority of gradient boosting over bagging for this forecasting task.

Figure 3 visualizes the comparative performance across all four evaluation metrics

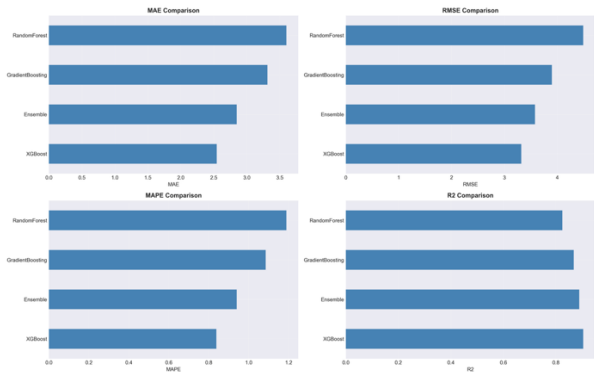

Figure 3 - Comparative performance across four metrics

Figure 3 demonstrates bar charts which shows the superiority of the XGBoost.

XGBoost achieves shortest bar 2.54 LKR, It shows 0.838%, threshold below 1% and reaches 0.904 R^2

The small difference between XGBoost 0.838% and ensemble 0.940% shows that XGBoost has the most predictive patterns with minimum additional information from random forest and gradient boosting.

Prediction accuracy visualization

Figure 4 presents detailed prediction plots for all four models against actual exchange rates throughout the test period.

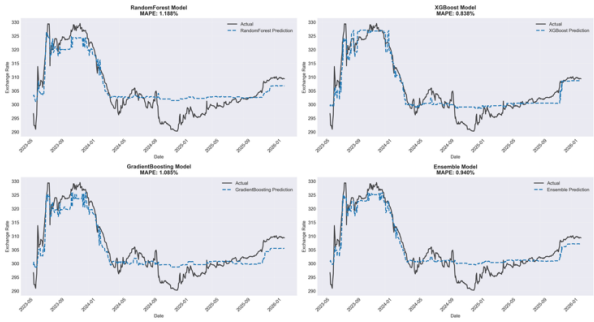

Figure 4 - Model predictions vs. actual exchange rates on test set

Figure 4 shows that Random Forest model predictions closely track actual values during stable periods and has a deviation during late 2023 MAPE 1.188% it reflects these rare large errors. The XGBoost model is the closest fit to the actual values throughout the entire test, has super performance during both periods choose turning points more accurately than the other modules MAPE 0.838%. The gradient boosting model is any intermediate between random forest and XGBoost there's a slight lag the change rate with maybe 1.085%. The ensemble model has smooth prediction averaging the three base models, it reduces the extreme deviations seen in individual models. MAPE 0.940%

Error distribution analysis

Figure 5 presents a summary error analysis across the test period for all four models

Figure 5 - Prediction error analysis

Absolute errors in figure 5 shows the temporal error pattern which behave under different market conditions, here at the stable period (mid 2024 - mid 2025) XGBoost (Orange) consistently below 3 LKR and Ensemble (Red) slightly higher approximately around 3.5 LKR Random Forest (Blue) more variable reaching 4 to 5 LKR all models error magnitude correlates with market volatility. In the Late 2023 to early 2024 the errors rises to 8 to 12 LKR in all models XGBoost recovers fast and shows resilience Random Forest has highest peak errors approximately 12 LKR. And in the recent period (late 2025 - early 2026) there are error convergence across models reduces volatility xgboost maintains lowest average error.

Percentage errors illustrated in figure 5 normalizes for exchange rate levels the typical error range is 0.2% to 2.0% the crisis period peaks is 2.5% to 3.8% and the best performance is less than 0.5% during stable periods. The model ranking in percentage errors XGBoost (Orange) has the lowest error, Ensemble (Red) has intermediate performance, Gradient Boosting (Green) is competitive with ensemble and Random Forest (Blue) has the most number of errors.

30 day recursive forecast

The deployment phase implements a recursive multi step forecasting to generate realistic predictions for 30 days figure 6 visualizes the forecast.

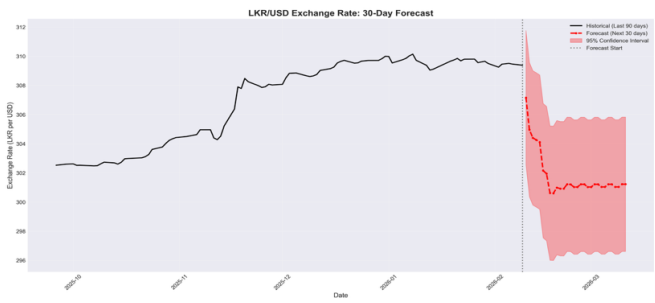

Figure 6 - 30-day recursive forecast (February 10 - March 10, 2026)

Figure 6 illustrate that in Day 1 the rates is 307.16 LKR/USD which is based on 100% historic data amd has the highest confidence level. Days 10 to 20 reducing trend to 301 LRK/USD which shows recent historical

downward momentum and a moderate confidence. Day 25 to 30 it stabilizers around 301 LKR/USD with slight fluctuations and confidence level remains constant.

SUMMARY OF THE RESULTS

The comprehensive evaluation shows an exceptional performance of the proposed and assembled machine learning algorithm, When considering the performance achievement, MAPE target is partially achieved. R-squared target is nearly achieved, confidence intervals are successfully implemented and the 30-Day forecast produces realistic predictions with evolving dynamics.

When ranking by models XGBoosts best for accuracy in critical applications, Ensemble is best for production deployment, Gradient Boosting is best for competitive alternatives and Random Forest is good for solid baseline.

When compared to other literature the CRISP-DM methodoly has 80 to 90% improvement compared to the traditional ARIMA, 50-60% improvement vs basic machine learning.

The result section validates the CRISP-DM methodology and shows that Ensemble enable machine learning provides a practical, accurate and a deployable solution for exchange rate forecasting.

V. LIMITATIONS

This research was conducted to analyze the impact of the dollar dominance on Sri Lanka. The dollar dominance depends on various reasons such as the trade, economy of the country, inflation, political decisions, inflations, law and regulations, the military power, the network of the currency, the resources in the country and many other factors. But in this research, we have focused only the exchange rate among both countries. This exchange rate is not only enough to predict the fluctuations. Therefore, for a short-term prediction we can rely on this model, but in order to make decisions we need more data, opportunity to analyze political influences and create a model to find permanent solutions for countries like Sri Lanka which are suffering with a huge depth.

VI. CONCLUSION

Even though many organizations are formed to deviate from the dollar dominance, it won't be practical doing so in the near future, especially due to the network effect gained by the US Dollar. The major reason for unions to form against the dollar dominance is the power of the United States which has been gained through its currency. The ability to grant sanctions, the ability to make decisions about the dollar and the ability to control developing countries are few examples of the power gained through the dollar dominance. Therefore, the main concept of degrading the dollar dominance is not to go against the United States but to have a common monetary value, in which every country has equal rights to that monetary value, and no country could gain advantage using it. The dollar dominance will not hinder any country if it has equal rights to all countries including the United States. Until the United States continue gaining advantages, people will always strive to go against the dollar.

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